Day : 11

* What is Malware Traffic Analysis?

- When malware infects a machine, it generates network traffic. As a DFIR analyst, your job is to read that traffic and reconstruct what happened.

* The Analyst Workflow — Always follow this order.

1) Protocol Hierarchy → What protocols exist?

2) Statistics → Conversations → who is talking to whom?

3) DNS Filter → What domains did it contact?

4) HTTP Filter → What URLs did it request?

5) Export HTTP objects → Were any files downloaded?

6) Check unusual ports → Port 4444? Port 1337? Suspicious!

* What Malware Traffic Looks Like

1) DNS Indicators :

Pattern	What it means
Random - looking	DGA - Domain Generation

domain	Algorithm
Same domain queried repeatedly	Beaconing
Long subdomain stings	DNS tunneling
Domains ending in .ru, .xyz, .tk	Often malicious

2) HTTP Indicators:

Pattern	What it means
POST to strange URL	Data exfiltration
GET of .exe or .dll	Malware ~~disguising itself~~ download
Ususual User-Agent	Malware disguishing itself
HTTP to IP (no domain)	Direct C2 connection

3) TCP Indicators:

Pattern	What it means
Connection to port 4444	Metasploit default
Connection to port 1337	Common malware port
Many SYN with no response	Port scanning
Regular timed connections	Beaconing

* Key Wireshark Techniques for Malware Analysis:

1) Protocol Hierarchy:

Statistics → Protocol Hierarchy

Shows every protocol in the PCAP as a percentage. Unexpected protocols = red flag.

2) Conversations:

Statistics → Conversations → TCP tab

Shows which IPs talked the most. Suspicious IP with lots of traffic = likely C2.

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3) Export HTTP Objects:

File → Export Objects → HTTP

Extracts all files transferred over HTTP — executables, scripts, documents.

4) Follow TCP Stream:

Right Click Packet → Follow → TCP Stream

Shows the full conversation between two hosts in human-readable form.

5) Check User-Agent:

Filter: http.user_agent

Malware often uses fake or unusual User-Agent strings.

* Common Malware Families and Their Traffic Patterns

Malware	Traffic Pattern
Emotet	HTTP POST to compromised WordPress sites
Trickbot	HTTPS to many IPs, certificate anomalies
Cobalt Strike	Beaconing every 60 seconds, HTTPS.

Drider	HTTP GET of encrypted payloads
Qakbot	HTTP + DNS with DGA domains

* Diagrams :

1) Malware PCAP analysis workflow

Step 1 — Protocol Hierarchy
Statistics → Protocol Hierarchy

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Step 2 — Conversations
Statistics → Conversations → TCP

↓

Step 3 — DNS Filter
Suspicious domains? DGA?

↓

Step 4 — HTTP Filter
POST requests? Suspicious URLs?

↓

Step 5 — Export Objects
File → Export Objects → HTTP

↓

Step 6 — Follow TCP Stream
Right Click → Follow → TCP Stream

2) Red flags to look for :

DGA Domain
xkq7z3mpa.ru — random-looking

Beaconing
Same IP every 60 seconds

HTTP POST Exfiltration
Data sent OUT to C2

Unusual Ports
Port 4444, 1337 = malware

Fake User-Agent
Malware disguising itself

HTTP to Bare IP
No domain = Direct C2

3) Key filters to malware analysis

dns	http
tcp.port == 4444	http.user_agent

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* Lab Observations:

- Findings:
- A modandcrackapk.com
- A wpad.nemotoads.health

→ Suspicious

Wireless Tools Help

Protocol	Lengtl	Info
DNS	80	Standard query 0xa0e2 A fonts.googleapis.com
DNS	80	Standard query 0xceac HTTPS fonts.googleapis.com
DNS	96	Standard query response 0xa0e2 A fonts.googleapis.com A 142.251.186.95
DNS	137	Standard query response 0xceac HTTPS fonts.googleapis.com SOA ns1.google.com
DNS	75	Standard query 0x78f6 A www.gstatic.com
DNS	75	Standard query 0x5222 HTTPS www.gstatic.com
DNS	100	Standard query response 0x5222 HTTPS www.gstatic.com HTTPS
DNS	91	Standard query response 0x78f6 A www.gstatic.com A 142.250.113.94
DNS	77	Standard query 0xfdf4 A fonts.gstatic.com
DNS	77	Standard query 0xed5d HTTPS fonts.gstatic.com
DNS	93	Standard query response 0xfdf4 A fonts.gstatic.com A 142.250.138.94
DNS	102	Standard query response 0xed5d HTTPS fonts.gstatic.com HTTPS
DNS	75	Standard query 0x1b23 A code.jquery.com
DNS	75	Standard query 0x2b26 HTTPS code.jquery.com
DNS	136	Standard query response 0x2b26 HTTPS code.jquery.com SOA george.ns.cloudflare.c
DNS	139	Standard query response 0x1b23 A code.jquery.com A 151.101.66.137 A 151.101.194.
DNS	81	Standard query 0x3c54 A wpad.nemotoads.health
DNS	157	Standard query response 0x3c54 No such name A wpad.nemotoads.health SOA nemotode
DNS	75	Standard query 0x7e8b A wpad.mshome.net
DNS	161	Standard query response 0x7e8b No such name A wpad.mshome.net SOA ns1-35.azure-d
DNS	72	Standard query 0xd975 A www.bing.com
DNS	72	Standard query 0xeee7 HTTPS www.bing.com
DNS	254	Standard query response 0xeee7 HTTPS www.bing.com CNAME www-www.bing.com.traffic
DNS	273	Standard query response 0xd975 A www.bing.com CNAME www-www.bing.com.trafficmana

Wireless Tools Help

Protocol	Lengtl	Info
DNS	159	Standard query response 0x2c32 A maps.googleapis.com A 142.250.115.95 A 142.251.186.95 A 142.250.1
DNS	133	Standard query response 0x10e5 HTTPS maps.gstatic.com SOA ns1.google.com
DNS	136	Standard query response 0xcbae HTTPS maps.googleapis.com SOA ns1.google.com
DNS	80	Standard query 0x22d1 A modandcrackedapk.com
DNS	80	Standard query 0x2182 HTTPS modandcrackedapk.com
DNS	96	Standard query response 0x22d1 A modandcrackedapk.com A 193.42.38.139
DNS	146	Standard query response 0x2182 HTTPS modandcrackedapk.com SOA ns1.dns-parking.com
DNS	78	Standard query 0x82df A edge.microsoft.com
DNS	78	Standard query 0x44e6 HTTPS edge.microsoft.com
DNS	181	Standard query response 0x82df A edge.microsoft.com CNAME edge-microsoft-com.dual-a-0036.a-msedge.
DNS	83	Standard query 0xdf20 A confirmsubscription.com
DNS	83	Standard query 0x7ade HTTPS confirmsubscription.com
DNS	115	Standard query response 0xdf20 A confirmsubscription.com A 13.56.30.207 A 52.8.34.0
DNS	139	Standard query response 0x7ade HTTPS confirmsubscription.com SOA ns0.dnsmadeeasy.com
DNS	149	Standard query response 0x44e6 HTTPS edge.microsoft.com CNAME edge-microsoft-com.dual-a-0036.a-mse
DNS	79	Standard query 0x059f A css.createsend1.com
DNS	79	Standard query 0xec30 HTTPS css.createsend1.com
DNS	78	Standard query 0xc101 A js.createsend1.com
DNS	78	Standard query 0xdca2 HTTPS js.createsend1.com
DNS	185	Standard query response 0xc101 A js.createsend1.com CNAME d33w6v2v5ta015.cloudfront.net A 18.160.1
DNS	186	Standard query response 0x059f A css.createsend1.com CNAME d33w6v2v5ta015.cloudfront.net A 18.160.
DNS	121	Standard query response 0xdca2 HTTPS js.createsend1.com CNAME d33w6v2v5ta015.cloudfront.net
DNS	122	Standard query response 0xec30 HTTPS css.createsend1.com CNAME d33w6v2v5ta015.cloudfront.net
DNS	78	Standard query 0x7138 A edge.microsoft.com